

Field Notes Journal #1: Why Earth Science?

The three ideas that resonated with me after watching the video "Why Earth Science" was the fact that Earth science influences many fields, from climate change, to natural disasters, and that we as humans should be stewards and take care of the Earth as best as we can to protect not only it, but us and the living organisms that live on the planet.

Another idea that resonated was how Earth science can explain many different phenomena, from erosion, to rock formation, that occurs over a period of many years.

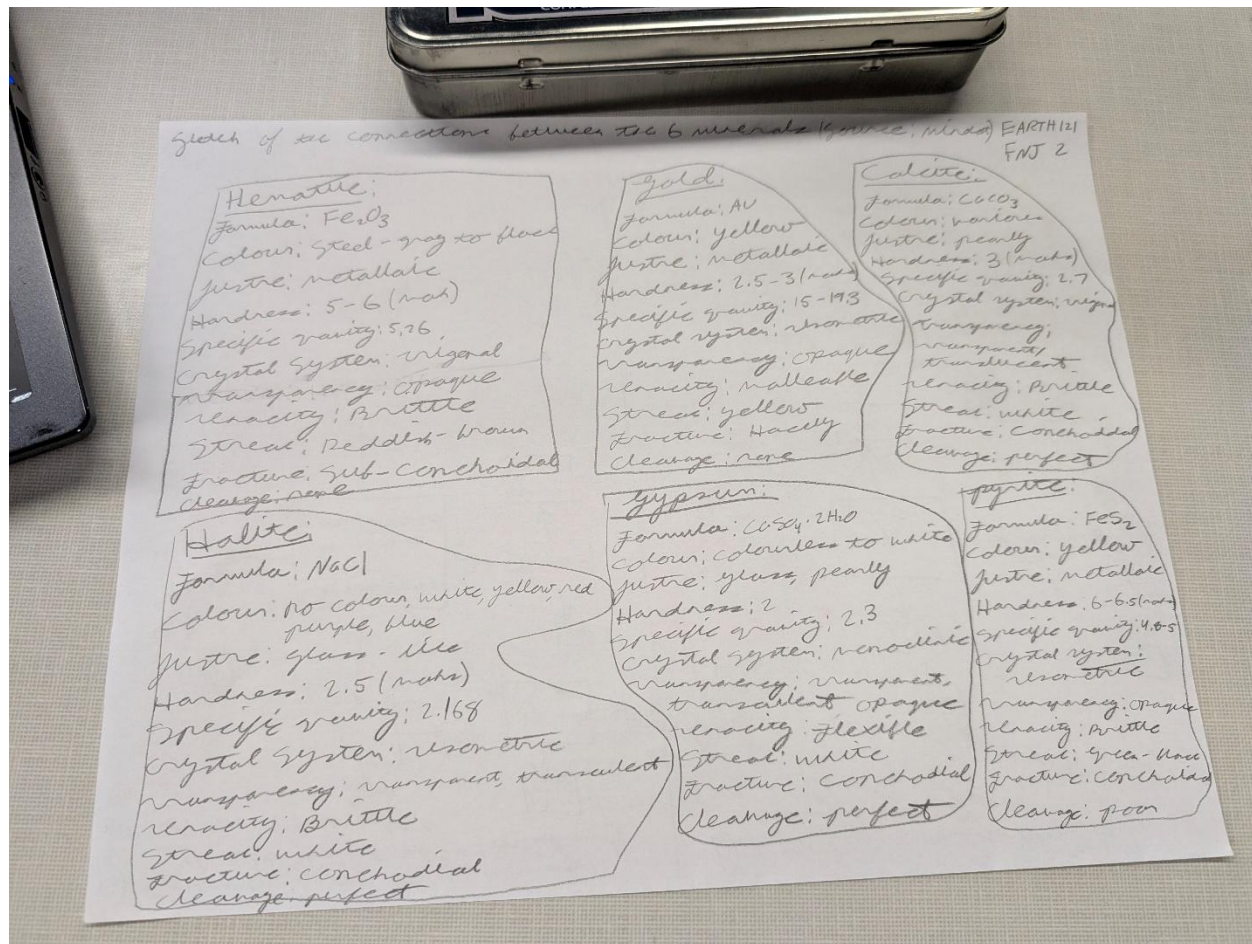
Finally, geoscientists use many different tools at their disposal to model the Earth, such as computer simulations, to see how different types of events affect different areas around the Earth.

What surprised me was the fact that Earth Science is a vast field that explains many things from Earth, from times where we weren't around, to predicting the future on what is going to happen to the Earth. What also surprised me was the fact that Earth Science encompasses many different components in the overall system of the Earth, from climate change to natural events, that it's vital to understand it to make informed decisions that can change the course of Earth.

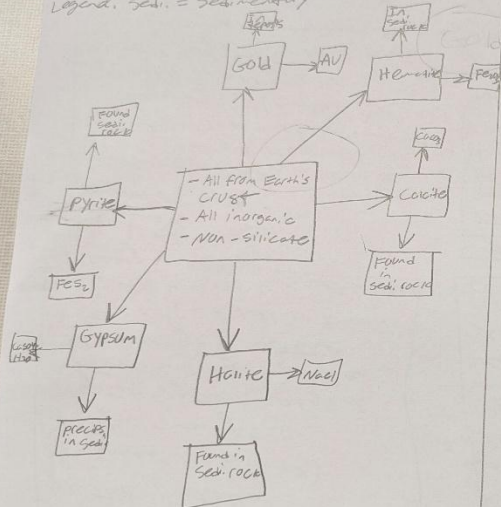
What changed the way I thought about the Earth was the fact that the Earth is ever evolving, and we are only a small piece of the puzzle in the grander scale of what the Earth does. From continents forming due to shifts in tectonic plates, to glaciers being created and then being destroyed due to factors like climate change, Earth is constantly changing, and even though we can influence those changes, we can't control everything that happens on Earth.

Field Notes Journal #2: Are Minerals Important?

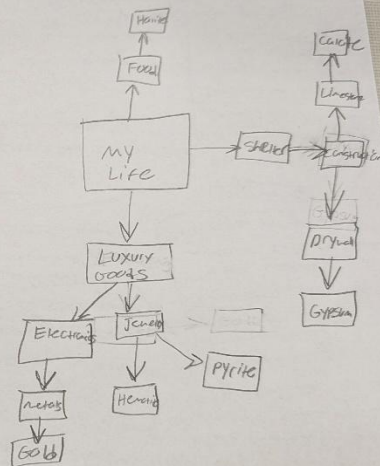
Images of Mind Map:



Legend: $\text{red}_1 = \text{red}_{\text{member}}$



b). Common connection to everyday life



Explanation of Mind Map:

For part A, I connected all of the minerals that were listed to the fact that there were three common connections between them: a). They all originated from the Earth's crust, either from sedimentary rocks or from deposits b). They are all inorganic minerals, as observed with their chemical formulas c). They are non-silicate, as they don't have that structure between them

For part B, I connected each mineral to everyday life in different ways, such as:

- *Halite* - used for food that I commonly eat, such as meats and nuts
- *Gypsum* - used for drywall, which is used in the construction of the various types of shelters that I use daily
- *Calcite* - used for limestone, which serves as a basis for buildings in construction which is used for shelters that I use on a various basis
- *Hematite/Pyrite* - used in jewelry, which are used in luxury goods that my family owns as heirlooms
- *Gold* - part of the precious metals used in my various pieces of technology, which would be under luxury goods as well

Field Notes Journal #3: What do you See in the Video?

What did I notice in the video?

What I noticed in the video was that initially, when the magma emerged from the volcano to become lava, it was runny and had a bright orange/red hue to it. However, continuing, it started to get very dark, and the consistency of the lava turned viscous in consistency, with air bubbles getting created and destroyed frequently. The lava also had a certain level of being uncontrollable, like a dam being broken and being overtaken by the river currents.

How does lava move across the Earth's surface?

Lava moves across the Earth with a viscous consistency about it. Lava isn't as free flowing as water but doesn't move as slow as molasses. Also, the flow is a bit analogous to a river, as in the video, it showed how the lava was sloshing about, with air bubbles forming and popping, launching the lava into the air in a sputter pattern.

Field Notes Journal #4: Is This Depiction of the Rock Cycle Complete and Accurate?

Images:

Figure 1

Simplified Rock Cycle diagram for kids, courtesy of ducksters

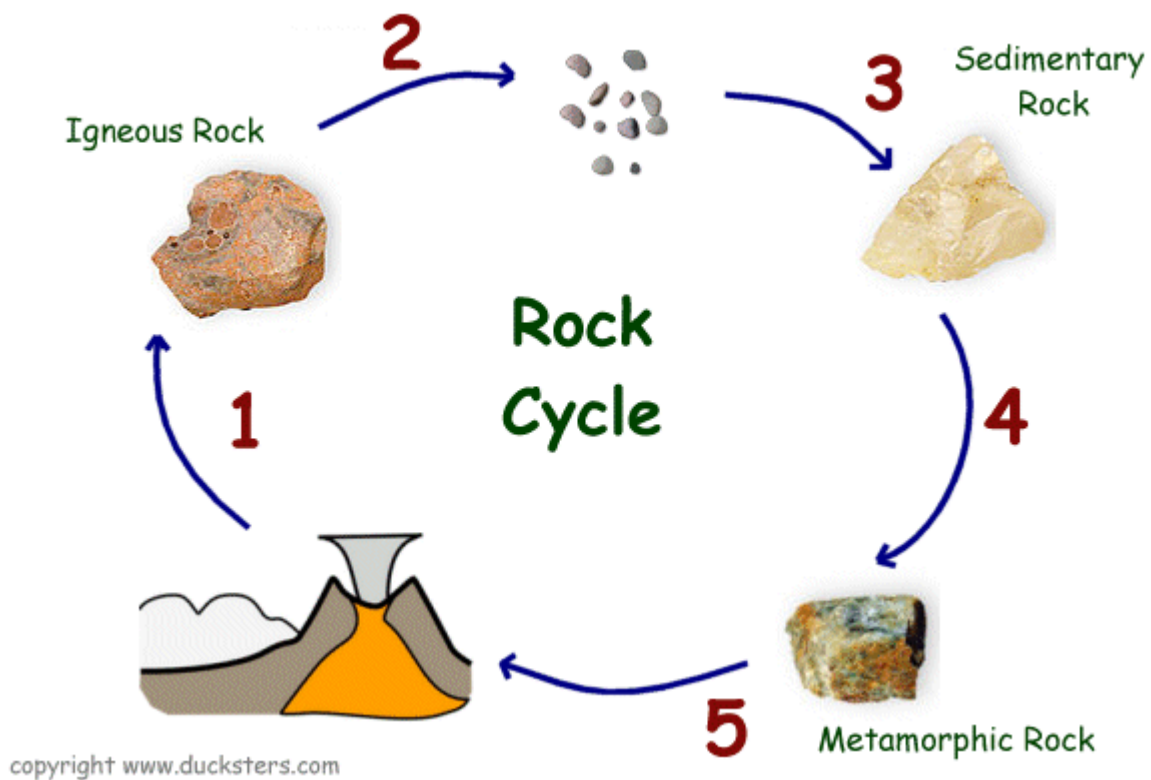
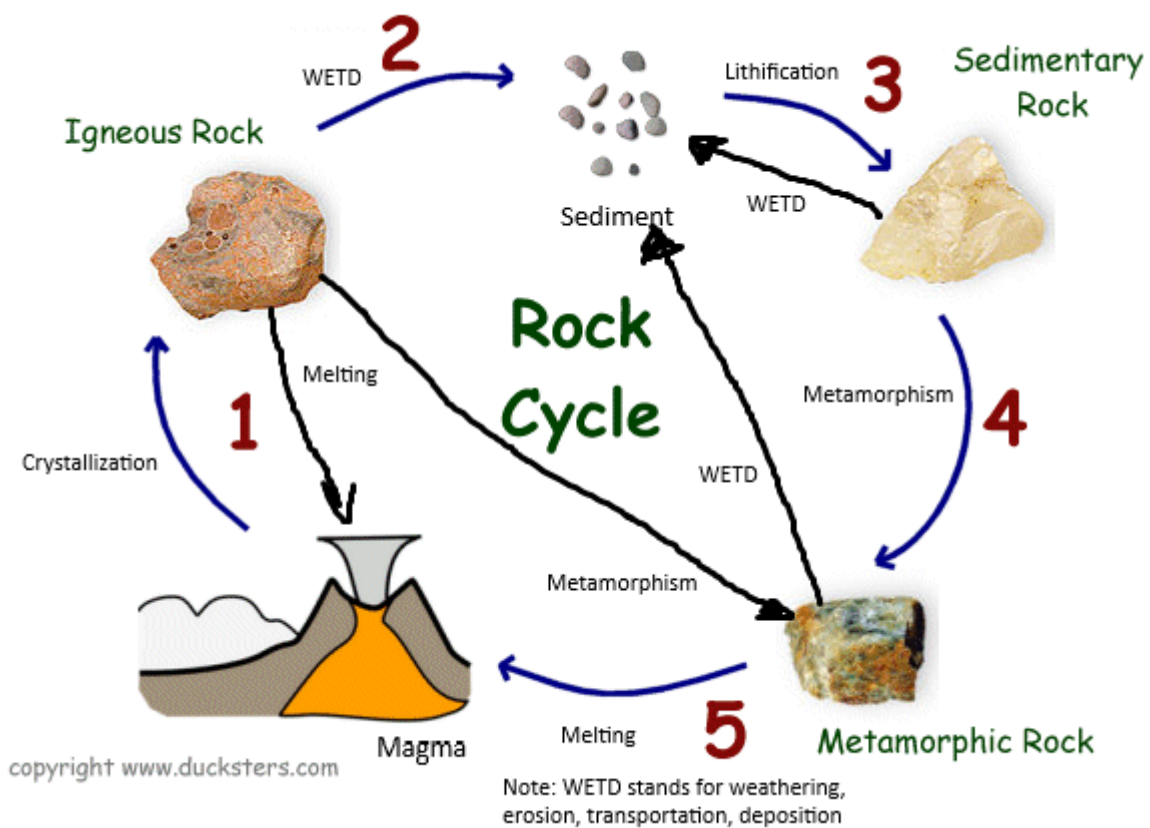


Figure 2

Annotated Rock Cycle diagram, using the above image as a basis



Impact of Incomplete Rock Cycle

The impact of an incomplete rock cycle for learning about the systems of the Earth is substantial. The original illustration was meant for children to learn about the rock cycle in a simplified manner. The problem with simplifying the rock cycle is that it loses the fact that the rock cycle is interconnected and has many parts that connect to one another. For example, in the original image, it shows the rock cycle starting from the magma phase and going to igneous rock, then going to sedimentary rock, and then metamorphic rock in that order. However, the rock cycle doesn't have specific steps and can jump between phases through different processes. This can lead children to think that the rock cycle is a rigid structure, but in reality, is quite flexible depending on the processes done, and they may think that Earth's systems are separate, but are the most interconnected systems that humans have described in the world. Also, there were no labels to describe each cycle (on the website, it was a brief description), which makes the original diagram of the rock cycle even more incomplete and not clear for children trying to learn it.

Field Notes Journal #5: Simplified Geologic Time Scale Chart

Supereon	Eon	Era	Period	Epoch	Age	Numerical Age
	Phanerozoic	Cenozoic	Quaternary	Holocene	Meghalayan	Present
					Northgrippian	4.2 ka
					Greenlandian	8.276 ka
				Pleistocene	Upper	11.7 ka
			Neogene	Pliocene	Piacenzian	2.58 Ma
				Miocene	Messinian	5.3 Ma
			Paleogene	Oligocene	Chattian	23 Ma
				Eocene	Priabonian	33.9 Ma
				Paleocene	Thanetian	56 Ma
		Mesozoic	Cretaceous	Upper	Maastrichtian	66 Ma
			Jurassic	Upper	Tithonian	145 Ma
			Triassic	Upper	Rhaetian	201 Ma
		Paleozoic	Permian	Lopingian	Changhsingian	252 Ma
			Carboniferous	Pennsylvanian	Gzhelian	299 Ma
			Devonian	Upper	Famennian	359 Ma
			Silurian	Pridoli		419 Ma
			Ordovician	Upper	Hirnantian	443 Ma
			Cambrian	Furongian	Stage 10	485 Ma
Precambrian	Proterozoic	Neo-proterozoic	Ediacaran			541 Ma

	Archean	Neo- archean				2.5 Ga
		Eoarchean				4.0 Ga
	Hadean					4.6 Ga